ENDPOINTHUB

Windows Endpoint Management System

Comprehensive Administrator, Deployment & Security Hardening Guide

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Target Platform: Windows 10, Windows 11 & Windows Server 2019/2022/2025 (x64)
System Scope: ASP.NET Core 9 Server Engine, React 19 Management Console, .NET 9 LocalSystem Agent
Classification: Enterprise IT Operational Runbook & Architecture Specification
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EXECUTIVE OVERVIEW: *EndpointHub is a unified, defense-in-depth Windows Endpoint Management platform designed to replace legacy Active Directory Group Policy Objects (GPOs) and complex MDM tools with a lightweight, real-time, tamper-resistant architecture. EndpointHub gives IT and Security administrators full administrative dominance over distributed Windows fleets, enforcing strict execution controls, zero-trust UAC restrictions, automated BitLocker recovery escrow, remote scripting, live performance telemetry, and machine lifecycle orchestration through an intuitive React web portal and native Windows installers.*

1. System Architecture & Administrative Security Model

EndpointHub employs an asymmetric, hub-and-spoke control plane where the server serves as the policy authority and telemetry aggregator, while distributed endpoints execute tasks under the highest local security context: **NT AUTHORITY\SYSTEM**.

1.1 High-Level Architecture Components

- **EndpointHub.Server:** A high-performance ASP.NET Core 9 Windows Service hosting REST endpoints, an internal Certificate Authority (CA) for device authentication, PostgreSQL persistence via Entity Framework Core 9, and a SignalR real-time duplex channel.
- **EndpointHub.Portal:** A modern, reactive web application built with React 19, TypeScript, and Vite. Embedded directly inside the server's wwwroot, providing unified fleet visibility, visual policy bundling, device grouping, live telemetry gauges, and an interactive remote shell.
- **EndpointHub.Agent:** A self-contained, single-file .NET 9 Windows Service deployed to endpoints. It operates with zero third-party dependencies, runs under LocalSystem, interacts directly with the Windows Registry, WMI/CIM, and the Windows API (e.g. GlobalMemoryStatusEx, NetRenameMachineInDomain), and connects over TLS with a rotating JWT; the client certificate is provisioned but the server does not enforce mTLS.

1.2 The Administrative Dominance & Privilege Model

A primary requirement of modern enterprise endpoint management is ensuring standard and local admin users cannot tamper with management software or bypass security policies. EndpointHub enforces strict hierarchy:

- **LocalSystem Elevation:** The agent runs as NT AUTHORITY\SYSTEM, allowing it to modify protected registry keys under HKLM\SOFTWARE\Policies, configure Microsoft Defender security preferences via PowerShell API, inspect BitLocker keys, and rename machines regardless of who is logged on.
- **Restricting Other App Admin Privileges:** EndpointHub actively prevents unprivileged applications and interactive local administrators from elevating silently or executing rogue code. By enforcing ConsentPromptBehaviorAdmin = 2 (prompt for credentials on secure desktop) and FilterAdministratorToken = 1, split-token admin sessions are held to strict scrutiny.
- **Anti-Tamper & Registry Integrity:** All policies pushed from the server are written to hardened system policy subkeys. Software Restriction Policies (SRP) permanently block unauthorized executable binaries in user-writable paths (e.g. %AppData%, %LocalAppData%\Temp), blocking ransomware and unsigned bypass scripts.

2. Server & Client Installation Blueprints

2.1 Server Installation (EndpointHubServerSetup.exe)

The EndpointHub Server is distributed as a native Inno Setup 6 installer (EndpointHubServerSetup.exe, ~42 MB). It packages the entire self-contained server binary, pre-built React portal, automated SSL/TLS generator, and service registration scripts.

Installation Steps:

1. Launch EndpointHubServerSetup.exe with Administrator privileges.
2. The setup wizard displays the **Configuration Dialog** requesting: Server Hostname/IP, HTTPS Port (default 443), Organization Name, Initial Admin Email, and Admin Password.

3. Setup automatically configures Windows Firewall rules, initializes directories (C:\EndpointHub\Server, Data, Logs, Backups), generates the internal CA and server certificates, and starts the EndpointHubServer Windows Service.
4. Once finished, access the Web Console at https://<Server_IP>:443.

2.2 Client Agent Installation (EndpointHubAgent.msi)

The EndpointHub Agent is packaged as an enterprise-grade Windows Installer MSI (EndpointHubAgent.msi, ~27.6 MB) built with WiX Toolset v4. It supports standard command-line parameters for zero-touch deployment.

MSI Command-Line Installation Syntax:

```
msiexec.exe /i EndpointHubAgent.msi SERVER_URL="https://mgmt.company.lan:443"
ENROLLMENT_CODE="A9F2B81C" /qn
```

Supported Deployment Tools:

- **Active Directory GPO:** Deploy via Computer Configuration → Software Settings → Software Installation.
- **Microsoft Intune:** Package as a Line-of-Business (LOB) or Win32 app (IntuneWin) passing the enrollment parameters.
- **PowerShell Mass Rollout:** Execute Install-Agent.ps1 -ServerUrl <URL> -EnrollmentCode <CODE> via WinRM or SCCM/MECM runbook.

3. Fleet Management, Grouping & Remote Control

3.1 Device Groups & Hierarchical Policy Assignment

Endpoints can be organized into logical groups (e.g., *Workstations*, *Finance Dept*, *Field Laptops*, *Servers*). Groups simplify policy administration and lifecycle workflows:

- **Color-Coded Visual Tags:** Each group features a custom hex color badge displayed throughout the web console and device inventory.
- **Group Policy Bundles:** Attaching one or more Policy Profiles to a Device Group automatically propagates effective policies to all enrolled member devices.
- **Policy Resolution Order:** Device-specific policies take precedence over Group policies, allowing surgical exceptions without duplicating entire policy profiles.

3.2 Zero-Touch Computer Renaming

EndpointHub supports seamless remote computer renaming directly from the portal without requiring domain admin credentials on the target machine. When an administrator initiates a rename:

1. The portal issues a RenameComputer command over SignalR to the agent.
2. The agent executes NetRenameMachineInDomain or WMI Rename() with zero user intervention.
3. The agent registers a pending reboot flag in HKLM\SOFTWARE\EndpointHub\Agent:PendingRename and notifies the server.
4. The portal flags the device with an amber status badge: *Renaming to: NEW-NAME (Reboot Pending)*.
5. Upon system restart, Windows updates the machine name, and the agent reports its new identity on its initial heartbeat.

3.3 BitLocker Key Escrow & Instant Recovery

Full-disk encryption keys are critical assets. The EndpointHub agent automatically inspects BitLocker status via manage-bde -protectors -get C:. If numerical password protectors exist, the agent safely transmits the 48-digit recovery password over the encrypted TLS channel for escrow in the server database. Administrators can inspect or copy the key from the Device Detail page with full audit logging.

3.4 Remote Script Orchestration (PowerShell / CMD)

Administrators can execute ad-hoc PowerShell or Command Prompt scripts remotely against any online device. The agent executes the script under the isolated LocalSystem environment with a configurable timeout (default 60s), capturing exit codes, standard output, and standard error for immediate inspection in the web console.

3.5 Live Hardware Telemetry & Process Surveillance

Endpoints stream live performance telemetry every 30 seconds: Overall CPU utilization percentage, Physical RAM (Total, Used, Free via Win32 GlobalMemoryStatusEx), Ready Disks capacity, Pending Reboot flags, and the top 6 memory-consuming processes.

4. Composable Policy Matrix: 30+ Preconfigured Restrictions

EndpointHub includes a comprehensive library of pick-and-choose enterprise restrictions organized into six functional security categories. Each setting is backed by authoritative Windows Registry keys, CSPs, or PowerShell security providers.

Category & Setting Name	Policy Key & Mechanism	Enforced Value & Security Impact
UAC Elevation Lockdown **Admin Privilege Control**	ConsentPromptBehaviorAdmin HKLM\...\Policies\System	**2 (Prompt on Secure Desktop)**. Prevents background malware from elevating silently.
Standard User Elevation Block **Admin Privilege Control**	ConsentPromptBehaviorUser HKLM\...\Policies\System	**0 (Automatically Deny)**. Standard users cannot elevate or invoke UAC password prompts.
Filter Administrator Token **Admin Privilege Control**	FilterAdministratorToken HKLM\...\Policies\System	**1 (Enabled)**. Forces built-in Admin accounts to operate in protected split-token mode.
Secure Desktop Enforcement **Admin Privilege Control**	PromptOnSecureDesktop HKLM\...\Policies\System	**1 (Enabled)**. Freezes foreground input, preventing UI spoofing attacks.
LUA Enforce Lockdown **Admin Privilege Control**	EnableLUA HKLM\...\Policies\System	**1 (Always Enabled)**. Blocks applications from disabling Windows User Account Control.
Block Executables in %Temp% **Execution Restriction**	SRP Disallow Temp Exes HKLM\...\Safer\CodeIdentifiers	**Disallowed (0)**. Blocks ransomware and dropper binaries from executing in user Temp.
Block Scripts in %AppData% **Execution Restriction**	SRP Disallow AppData HKLM\...\Safer\CodeIdentifiers	**Disallowed (0)**. Prevents unauthorized .exe, .vbs, .bat, and .ps1 from AppData.
Disable Windows Installer (MSI) **Execution Restriction**	DisableMSI HKLM\...\Policies\MSI	**2 (Always Disabled)**. Completely prevents users from installing MSI packages.
Block Command Prompt (cmd.exe) **System Tool Lockdown**	DisableCMD HKCU/HKLM\...\Policies\System	**1 or 2**. Prevents interactive users from launching cmd.exe or running batch scripts.
Block Registry Editor (regedit) **System Tool Lockdown**	DisableRegistryTools HKCU/HKLM\...\Policies\System	**1 (Enabled)**. Blocks interactive users from inspecting or modifying the registry.
Block Task Manager **System Tool Lockdown**	DisableTaskMgr HKCU/HKLM\...\Policies\System	**1 (Enabled)**. Prevents users from viewing or killing system security services.
Real-Time Antivirus Protection **Microsoft Defender**	DisableRealtimeMonitoring HKLM\...\Windows Defender\Real-Time	**0 (Active)**. Ensures real-time scanning cannot be deactivated by local users.
Cloud-Delivered Protection **Microsoft Defender**	MAPSReporting HKLM\...\Windows Defender\Spynet	**2 (Advanced MAPS)**. Sends suspicious samples to Microsoft Cloud AI for analysis.
PUA / Adware Protection **Microsoft Defender**	PUAProtection HKLM\...\Windows Defender	**1 (Enabled)**. Blocks potentially unwanted applications, miners, and adware.
Controlled Folder Access **Anti-Ransomware**	EnableControlledFolderAccess HKLM\...\Windows Defender\Exploit	**1 (Lockdown)**. Blocks unauthorized apps from altering user documents.
Script Scanning & Inspection **Microsoft Defender**	DisableScriptScanning HKLM\...\Windows Defender\Real-Time	**0 (Scan All)**. Forces real-time inspection of VBScript, JScript, and PowerShell.
Total USB Storage Block **Data Loss Prevention**	Deny_All / UsbStor Start HKLM\...\StorageDevicePolicies	**Total Block (Start=4)**. USB mass storage devices cannot be mounted or accessed.
USB Read-Only Enforcement **Data Loss Prevention**	WriteProtect = 1 HKLM\...\StorageDevicePolicies	**Read-Only**. Users may read USB drives but cannot exfiltrate corporate data to them.
Optical Drive Lockdown (CD/DVD) **Data Loss Prevention**	Deny_All (CDROM) HKLM\...\RemovableStorageDevices	**Blocked**. Disables read/write access to internal and USB CD/DVD burners.

Category & Setting Name	Policy Key & Mechanism	Enforced Value & Security Impact
Portable Devices (WPD/MTP) **Data Loss Prevention**	Deny_All (WPD) HKLM\...\RemovableStorageDevices	**Blocked**. Prevents connecting Android/iOS phones as USB storage.
Domain Profile Firewall **Host Firewall**	EnableFirewall HKLM\...\FirewallPolicy\DomainProfile	**1 (Enforced)**. Blocks inbound connections not explicitly allowed by admin rules.
Private Profile Firewall **Host Firewall**	EnableFirewall HKLM\...\FirewallPolicy\StandardProfile	**1 (Enforced)**. Protects endpoints on home/office networks.
Public Profile Firewall **Host Firewall**	EnableFirewall HKLM\...\FirewallPolicy\PublicProfile	**1 (Maximum Defense)**. Drops all inbound traffic and stealth probes on public Wi-Fi.
Remote Desktop (RDP) Policy **Remote Access**	fDenyTSConnections HKLM\...\Terminal Server	**1 (Disabled) or 0 (NLA Required)**. Prevents unauthorized RDP access.
BitLocker Full-Disk Encryption **Endpoint Encryption**	EncryptionMethodWithXtsAes256 HKLM\...\FVE	**XTS-AES 256-bit**. Enforces hardware TPM encryption on all system drives.
Windows Update Enforcement **Patch Management**	AUOptions = 4 HKLM\...\WindowsUpdate\AU	**Auto-Install**. Downloads and installs security patches automatically.
No Auto-Reboot with Logged Users **Patch Management**	NoAutoRebootWithLoggedOnUsers HKLM\...\WindowsUpdate\AU	**1 (Enabled)**. Prevents unexpected mid-day reboot disruptions for active users.
Telemetry Data Cap **Privacy & Compliance**	AllowTelemetry = 0 HKLM\...\Policies\DataCollection	**Security / Minimum**. Restricts Windows diagnostic telemetry to critical security only.

5. Operational Runbook, Backup & Disaster Recovery

5.1 Database & Server Backup Procedures

EndpointHub includes an automated database backup subsystem. Backups are stored under C:\EndpointHub\Backups\. Administrators can trigger on-demand backups or restore previous snapshots directly from the **Backups** page in the web console.

Manual PostgreSQL Backup Command:

```
pg_dump.exe -U endpointhub -h localhost -F c -b -v -f "C:\EndpointHub\Backups\endpointhub_manual.dump" endpointhub
```

5.2 Certificate Lifecycle & Agent Authentication

Each agent connects over TLS and authenticates with a 30-day JWT that is rotated every 7 days; the client certificate is provisioned but server-side mTLS is not enforced. On initial enrollment, the agent generates an RSA 2048-bit key pair, submits a Certificate Signing Request (CSR) to the server, and imports the signed client certificate into Cert:\LocalMachine\My. The server's internal CA certificate is imported into Cert:\LocalMachine\Root. The installer pins the CA thumbprint by default. When CA or server certificates near expiry (within 30 days), the portal displays a prominent security banner on the dashboard.

5.3 Agent Diagnostics & Troubleshooting Runbook

When diagnosing endpoint connectivity or policy compliance issues, follow these verification steps:

1. Verify Windows Service Status:

Get-Service EndpointHubAgent (Must report *Running* with *Automatic* startup).

2. Inspect Agent Event Logs & Disk Logs:

Logs are continuously written to C:\ProgramData\EndpointHub\Agent\Logs\agent-*.log. Use PowerShell to inspect recent failures:

```
Get-Content "C:\ProgramData\EndpointHub\Agent\Logs\agent-*.log" -Tail 50
```

3. Inspect Registry State:

Inspect HKLM\SOFTWARE\EndpointHub\Agent. Confirm DeviceId, ServerUrl, and CertificateThumbprint are present.

4. Test Network & TLS Port:

Test TCP connectivity from endpoint to server port 443:

```
Test-NetConnection -ComputerName -Port 443
```

5.4 Clean Agent Uninstallation

To decommission or remove the agent, run the following elevated PowerShell command or invoke msiexec /x EndpointHubAgent.msi /qn:

```
Stop-Service EndpointHubAgent -Force -ErrorAction SilentlyContinue sc.exe delete EndpointHubAgent
Remove-Item "C:\EndpointHub\Agent" -Recurse -Force -ErrorAction SilentlyContinue Remove-Item "HKLM:\SOFTWARE\EndpointHub" -Recurse -Force -ErrorAction SilentlyContinue
```